

Non-Abelian gauge theories

1. Recap of gauge theory and brief look at Lie group theory

1.1 Recap: Abelian gauge theory

We firstly start with QED, which as we know is an Abelian $U(1)$ gauge theory. It start from the free fermion Lagrangian:

$$\mathcal{L} = i\bar{\psi}\gamma^\mu\partial_\mu\psi - m\bar{\psi}\psi. \quad (3.1)$$

The Noether current is: $j^\mu = \bar{\psi}\gamma^\mu\psi$. To approach the QED Lagrangian, we introduce the (global) gauge transformation as:

$$\psi \rightarrow \psi' = e^{i\alpha}\psi \quad (3.2)$$

$$\bar{\psi} \rightarrow \bar{\psi}' = e^{-i\alpha}\bar{\psi} \quad , \quad \alpha \in \mathbb{R}. \quad (3.3)$$

The local gauge transformation is just changing the parameter α rely on the coordinate:

$$\alpha \rightarrow \alpha(x). \quad (3.4)$$

The gauge symmetry is saying that the Lagrangian is invariant under the gauge transformation. This requires us to introduce the covariant derivative D_μ :

$$D_\mu\psi(x) = (\partial_\mu - igA_\mu)\psi(x). \quad (3.5)$$

where in QED case, the charge $g = e$, and the gauge field A_μ is just the photon field, which satisfies the transformation rule:

$$A_\mu \rightarrow A'_\mu = A_\mu + \frac{1}{g}\partial_\mu\alpha(x). \quad (3.6)$$

Then we immediately have the covariant derivative transfers as

$$D_\mu\psi \rightarrow (D_\mu\psi)' = \partial_\mu\psi' - igA'_\mu\psi' = e^{i\alpha}D_\mu\psi. \quad (3.7)$$

Thus the invariant of $\mathcal{L}$ requires that

$$\mathcal{L}(\psi, A_\mu) = \mathcal{L}(\psi', A'_\mu) = i\bar{\psi}\gamma^\mu D_\mu\psi - m\bar{\psi}\psi \quad (3.8)$$

$$= i\bar{\psi}\gamma^\mu\partial_\mu\psi - m\bar{\psi}\psi + gA_\mu j^\mu. \quad (3.9)$$

For photon dynamical, we need to add kinetic term for A_μ :

$$\mathcal{L}_{\text{Maxwell}} = -\frac{1}{4}F_{\mu\nu}F^{\mu\nu}. \quad (3.10)$$

with $F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu$.

1.2 $SU(N)$ gauge theory

Now we consider the non-Abelian $SU(N)$ gauge theory with N Dirac spinors ψ_j , where $j = 1, \dots, N$. The original Lagrangian is

$$\mathcal{L} = \sum_{j=1}^N i\bar{\psi}_j \gamma^\mu \partial_\mu \psi_j - m\bar{\psi}_j \psi_j. \quad (3.11)$$

Under the global unitary transformation $U_{ij} \in \mathbb{C}$, $UU^\dagger = \mathbb{1}$, the spinors obey

$$\psi_j \rightarrow \psi'_j = \sum_k U_{jk} \psi_k, \quad (3.12)$$

$$\bar{\psi}_j \rightarrow \bar{\psi}'_j = \sum_k \bar{\psi}_k U_{kj}^\dagger, \quad (3.13)$$

thus the mass term transformation under $SU(N)$

$$\sum_j \bar{\psi}_j \psi_j \rightarrow \sum_j \bar{\psi}'_j \psi'_j = \sum_{k,l,j} \bar{\psi}_k \underbrace{U_{kj}^\dagger U_{jl}}_{\delta_{kl}} \psi_l = \sum_k \bar{\psi}_k \psi_k \quad (3.14)$$

is invariant.

Similar with the Abelian case, the local gauge transformation is $U \rightarrow U(x)$. We also require the Lagrangian invariant under the transformation. This gives coupling of fermions to gauge field in covariant derivative analogous to QED:

$$D_\mu \psi_j = \partial_\mu \psi_j - ig \sum_k (A_\mu)_{jk} \psi_k. \quad (3.15)$$

This form of covariant derivative also requires the transformation rule as

$$D_\mu \psi_j \rightarrow (D_\mu \psi_j)' = \partial_\mu \psi'_j - ig \sum_k (A'_\mu)_{jk} \psi'_k \quad (3.16)$$

$$= \sum_k U_{jk} (D_\mu \psi_k) \quad (3.17)$$

For convenient, we will use the short-hand notation (Matrix notation), which basically just ignore the index of the matrix element:

$$D_\mu \psi \rightarrow (D_\mu \psi)' = U D_\mu \psi = e^{i\alpha(x)} D_\mu \psi. \quad (3.18)$$

We claim that the gauge field transforms following

$$A_\mu \rightarrow A'_\mu = U A_\mu U^\dagger - \frac{i}{g} (\partial_\mu U) U^\dagger. \quad (3.19)$$

Although we can directly use the definition to compute this expression, I just don't want to write so much. Instead, we can have a quick check that this claim indeed satisfies our requirement:

$$D_\mu \psi \rightarrow (D_\mu \psi)' = \partial_\mu (U\psi) - ig \left(U A_\mu U^\dagger - \frac{i}{g} (\partial_\mu U) U^\dagger \right) U\psi \quad (3.20)$$

$$= U \partial_\mu \psi + (\partial_\mu U) \psi - ig U A_\mu \psi - (\partial_\mu U) \psi \quad (3.21)$$

$$= U (\partial_\mu \psi - ig A_\mu \psi) = U D_\mu \psi. \quad (3.22)$$

Thus, the Lagrangian is invariant under $SU(N)$ gauge transformation.

Now we need to approach the dynamical of non-Abelian gauge field, which requires a field strength. Since we keep calling it as “non-Abelian”, but until now it’s not quite obvious to see how it is “non-Abelian”, therefore, the field strength should give you some direct idea when we claim:

$$F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu - i g [A_\mu, A_\nu]. \quad (3.23)$$

We can also check it is true that this field strength indeed satisfies our requirement, namely keep the Lagrangian invariant. To show this, we firstly have the transformation rule:

$$F_{\mu\nu} \rightarrow F'_{\mu\nu} = \partial_\mu A'_\nu - \partial_\nu A'_\mu - i g [A'_\mu, A'_\nu] = U F_{\mu\nu} U^\dagger. \quad (3.24)$$

The Lagrangian is proportion to the trace over $F_{\mu\nu} F^{\mu\nu} = (F_{\mu\nu})_{jk} (F^{\mu\nu})^{kj} = \text{tr}(F_{\mu\nu} F^{\mu\nu})$, due to the Lagrangian is a scalar, thus the only possible combination is the trace over $F_{\mu\nu}$. Thus $\text{tr} F_{\mu\nu}$ transfers as

$$\begin{aligned} \text{tr}(F_{\mu\nu} F^{\mu\nu}) &\rightarrow \text{tr}(F'_{\mu\nu} F'^{\mu\nu}) = \text{tr}(U F_{\mu\nu} U^\dagger U F^{\mu\nu} U^\dagger) \\ &= \text{tr}(U U^\dagger F_{\mu\nu} F^{\mu\nu}) \\ &= \text{tr}(F_{\mu\nu} F^{\mu\nu}). \end{aligned} \quad (3.25)$$

Then we get **the Lagrangian of Yang-Mills theory**

$$\mathcal{L} = -\kappa \text{tr}(F_{\mu\nu} F^{\mu\nu}) + \sum_{j,k=1}^N [\bar{\psi}_j \gamma^\mu (D_\mu)_{jk} \psi_k - m \bar{\psi}_j \psi_j], \quad (3.26)$$

where κ is normalized constant.

1.3 Recap for essential Lie group theory

To continue our journey, it’s necessary to master some basic knowledge about Lie group and Lie algebra. So in the next, we will recall some essential knowledge about these mathematical tools and show how are they used in gauge theory.

Firstly, we recall the infinitesimal transformation of the gauge operation. For Abelian theory, e.g. QED:

$$e^{i\alpha} \simeq 1 + i\alpha. \quad (3.27)$$

For non-Abelian gauge theory, e.g. Yang Mills theory:

$$U = \mathbb{1} + \hat{H} + \mathcal{O}(\hat{H}^2), \quad (3.28)$$

$$U^\dagger = \mathbb{1} + \hat{H}^\dagger + \mathcal{O}(\hat{H}^2). \quad (3.29)$$

Thus we have

$$U^{-1} = U^\dagger \implies \hat{H} = -\hat{H}^\dagger \quad (\text{anti-hermitian}) \quad (3.30)$$

where we note $\hat{H} = iH$ with $H = H^\dagger$. Expand $\hat{H}$ in parameters of transformation α^a

$$\hat{H}_{jk} = i \sum_a \alpha^a (t^a)_{jk} \quad (3.31)$$

with $\alpha^a \in \mathbb{R}$, $t^a = (t^a)^\dagger$ is hermitian. Thus t^a span a basis in the space of hermitian $N \times N$ matrices such that we can expand $\hat{H}$ in terms of this basis. t^a are called the generators of gauge transformations.

Generally, the unitary $N \times N$ matrix has N^2 linear independent generators due to t^a is $N \times N$. However, in our case, we require $\det U = 1$, which gives one constraint. So there are $\mathbf{N}^2 - \mathbf{1}$ generators in $\mathbf{SU}(\mathbf{N})$. Further more, base on $\det U = 1$ and the Jacobi identity $\det(e^M) = e^{\text{tr}(M)}$, we find

$$\det U = \det(e^{\hat{H}}) = e^{\text{tr} \hat{H}} = 1, \quad (3.32)$$

thus, $\text{tr} \hat{H} = 0$ and implies

$$\text{tr}(t^a) = 0. \quad (3.33)$$

So t^a are traceless generators.

Now we discuss the algebra structure for generators t^a . The Lie algebra, namely the commutator is given by:

$$[t^a, t^b] = i \sum_c f^{abc} t^c. \quad (3.34)$$

where f^{abc} is the structure constant of Lie algebra. The elements of Lie group can be expanded around $\mathbb{1}$.

$$g(\alpha) = \mathbb{1} + i \sum_{a=1}^d \alpha_a t^a + \frac{1}{2} \sum_{a,b=1}^d \alpha_a \alpha_b T^{ab} + \mathcal{O}(\alpha^3), \quad (3.35)$$

where $T^{ab} = T^{ba}$ is symmetric, which can be expanded in terms of the generator t^a, t^b . Notice $g(0) = \mathbb{1}$ and because of $g(\alpha)\mathbb{1} = \mathbb{1}g(\alpha)$, we then have

$$\gamma_a(\alpha_1, \dots, \alpha_d; 0, \dots, 0) = \gamma_a(0, \dots, 0; \alpha_1, \dots, \alpha_d) = \alpha_a \quad (3.36)$$

where the γ is defined by the group multiplication:

$$g(\alpha_1, \dots, \alpha_d) \cdot g(\beta_1, \dots, \beta_d) = g(\gamma_1, \dots, \gamma_d), \quad (3.37)$$

in which γ is the function of α and β :

$$\gamma_a = \gamma_a(\alpha_1, \dots, \alpha_d; \beta_1, \dots, \beta_d). \quad (3.38)$$

Base on the fact that $\gamma(\alpha, 0) = \alpha$ and $\gamma(0, \beta) = \beta$, then we have only one choice to expand γ_a until the second order to keep these relation, namely:

$$\gamma_a = \alpha_a + \beta_a + \sum_{b,c} C_{bc}^a \alpha_b \beta_c + \dots, \quad (3.39)$$

where C_{bc}^a is a unknown coefficient. Thus, use $g(\alpha)g(\beta) = g(\gamma)$, we have for the left-hand side:

$$g(\alpha)g(\beta) = \mathbb{1} + i \sum_a (\alpha_a + \beta_a) t^a + \frac{1}{2} \sum_{a,b} [(\alpha_a \alpha_b + \beta_a \beta_b) T^{ab} - 2\alpha^a \beta^b t^a t^b] + \dots, \quad (3.40)$$

And for the right-hand side:

$$g(\gamma) = \mathbb{1} + i \sum_a \gamma_a t^a + \frac{1}{2} \sum_{a,b} \gamma_a \gamma_b T^{ab} + \dots \quad (3.41)$$

Thus use (3.39), then the right-hand side becomes

$$g(\gamma(\alpha, \beta)) = \mathbb{1} + i \sum_a (\alpha_a + \beta_a) t^a + i \sum_{a,b,c} C_{bc}^a \alpha_b \beta_c t^a + \frac{1}{2} \sum_{a,b} (\alpha_a + \beta_a)(\alpha_b + \beta_b) T^{ab} + \dots \quad (3.42)$$

Compare (3.40) with (3.42), we find

$$-t^a t^b = i \sum_c C_c^{ab} t^c. \quad (3.43)$$

this gives the structure constant:

$$[t^a, t^b] = i \sum_{c=1}^d (C_c^{ba} - C_c^{ab}) t^c \quad (3.44)$$

$$= i \sum_{c=1}^d f^{abc} t^c \quad (3.45)$$

where $f^{abc} = -f^{bac}$, and d is the dimension of Lie algebra.